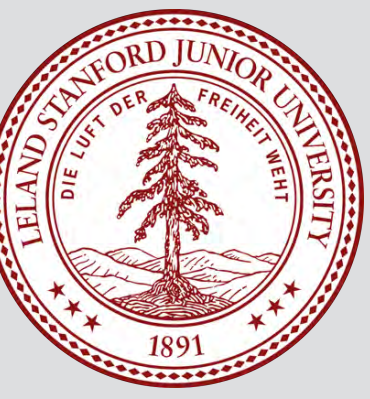


# Antidote Allocation for Epidemiological Models

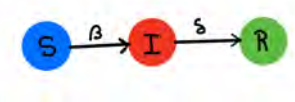
Uma Dayal, Senem Isik, Margalit Glasgow, Mary Wootters

Stanford University

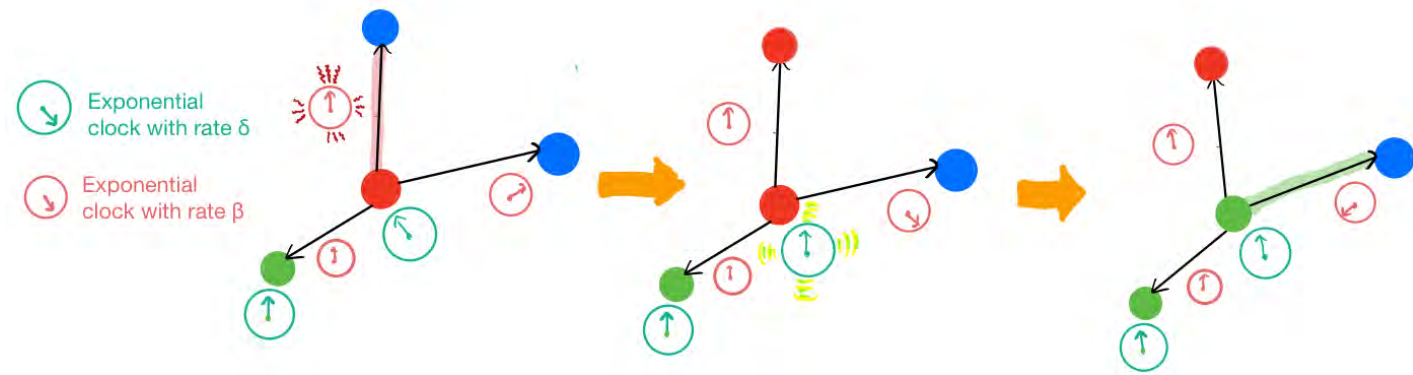


## Background

### SIR Model on Networks



Each node in the network is one of three states Susceptible (S), Infected (I), or Recovered (R). The rate at which a node gets infected is called the **transmission rate** ( $\beta$ ). The rate at which a node recovers is called the **recovery rate** ( $\delta$ ).



### Challenge!

Given a finite budget of antidote, how do we distribute antidote to each node in a network to minimize the size of the infection?

### Antidote Allocation

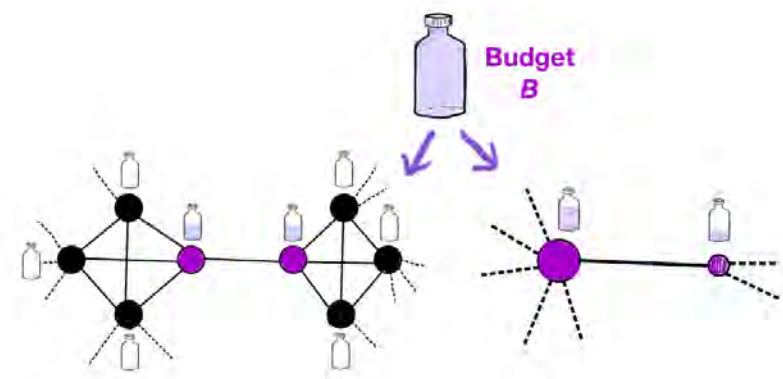


Figure 1: Left: a barbell graph where the antidote is uniformly given to the boundary nodes. Right: A high-degree and low-degree node where proportionally more antidote is given to the high-degree node.

We need to find a function of the budget  $f : \mathbb{R} \rightarrow \mathbb{R}^n$ , where  $n$  is the number of nodes, such that

- Objective:  $\min E[\text{size of infection}]$
- Constraint:  $\sum f(B)_i \leq B$  where  $f(B)_i$  is the amount of antidote given to node  $i$ .

Each unit of antidote increases the recovery rate of the node by a unit:  $f(B)_i = \delta_i$ , the recovery rate of node  $i$ . We get the constraint  $\sum \delta_i \leq B$ .

### Some differential equations...

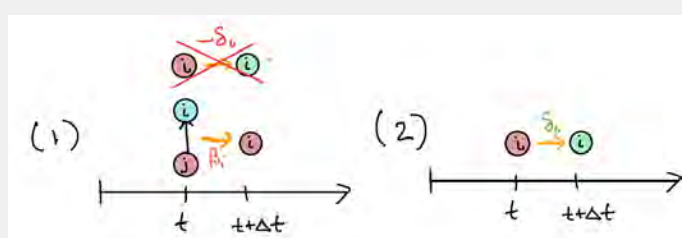
In an **adjacency matrix**  $A$ , we have  $A_{ij}$  is 1 if there is an edge between nodes  $i$  and  $j$  and 0 if there is no edge.

The differential equations for SIR on networks:

$$\frac{d}{dt}E[I_i(t)] = \beta \sum_{j=1}^n A_{ij}E[I_j(t)S_i(t)] - \delta_i E[I_i(t)] \quad (1)$$

$$\frac{d}{dt}E[R_i(t)] = \delta_i E[I_i(t)] \quad (2)$$

where  $I(t), S(t), R(t) \in \{0, 1\}^n$  are vectors designating whether each node is infected, susceptible or recovered, respectively.



## When does an epidemic occur?

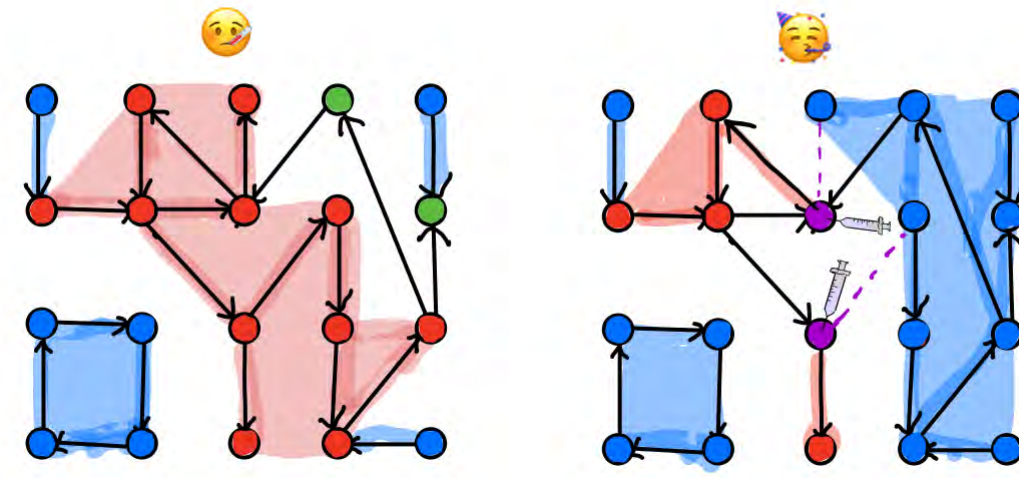


Figure 2: Left: an epidemic, Right: distributing antidote to reduce outbreak

### More differential equations (now with matrices)...

We define the matrices:

$A$  = adjacency matrix of the graph  
 $D$  = diagonal matrix where  $D_{ii} = \delta_i$

Because we know the number of susceptible nodes is always less than the number of initially susceptible nodes, equation (1) can be written as the inequality,

$$\frac{d}{dt}E[I_i(t)] \leq S_i(0)\beta \sum_{j=1}^n A_{ij}E[I_j(t)] - \delta_i E[I_i(t)] \quad (3)$$

This can be now written in matrix form:

$$\frac{d}{dt}E[I(t)] \leq (\beta A - D)E[I(t)]$$

which we can further solve to get

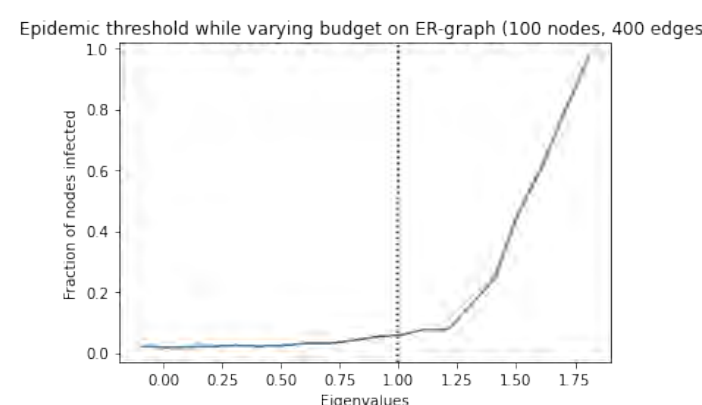
$$E[I(t)] \leq \exp((\beta A - D)t)I(0). \quad (4)$$

Similarly, (2) can be written as

$$\frac{d}{dt}E[R(t)] = DE[I(t)] \quad (5)$$

$$\Rightarrow \lim_{t \rightarrow \infty} E[R(t)] = D \int_0^\infty E[I(t)]dt \quad (6)$$

When top eigenvalue of the matrix  $\beta A - D$  is less than 1, then epidemic is unlikely but if above 1, epidemic occurs with high probability!



### Convex Program (in [2])

To minimize the number of nodes that eventually get infected we can minimize the RHS of equation (6) which can be combined with (4):  $D \int_0^\infty E[I(t)]dt \leq D \int_0^\infty \exp((\beta A - D)t)I(0)dt$ .

The **objective** is:

$$\min_D D \int_0^\infty \exp((\beta A - D)t)E[I(0)]dt \quad (7)$$

given the **constraint**:

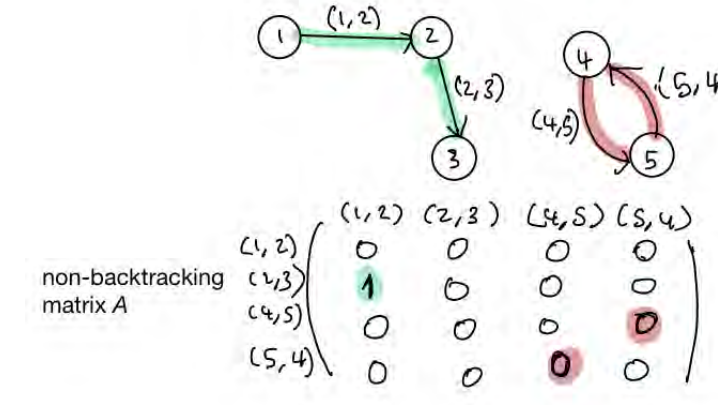
$$\sum_{i=1}^n \delta_i \leq B \quad (8)$$

**Note!** The above program isn't convex but can be converted into one.

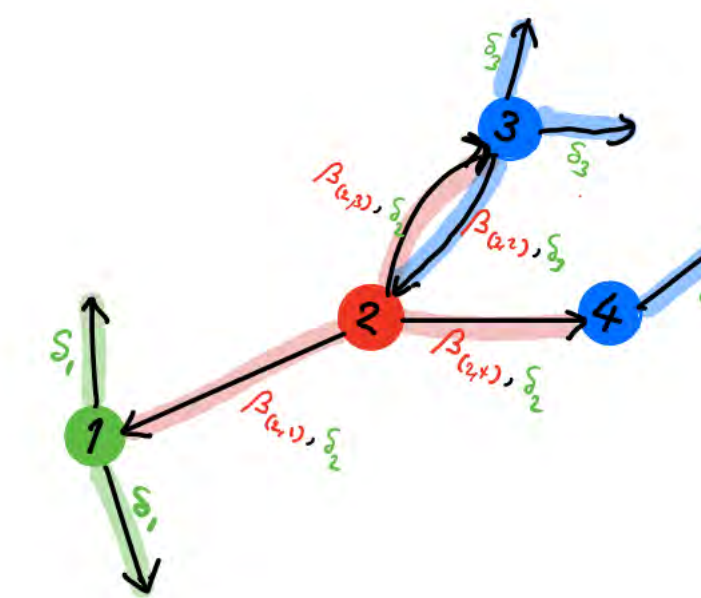
## Directed Edge Approach

The inequality in equation (3) is not very tight because it allows for 2-cycles which over-count the number of infected nodes.

To solve this issue, we generate a convex program in terms of directed edges, for which we use a **non-backtracking matrix**!



What does it mean to infect directed edges?



### Differential equations (now for directed edges)...

We have to define the new matrices (now indexed by directed edges):

$A$  = the non-backtracking matrix

$$D_{(i,j),(i,j)} = \delta_i$$

$$x_{(i,j)} = \frac{1}{deg(i)}$$

We can now rewrite the differential equation:

$$\begin{aligned} \frac{d}{dt}E[I_{(i,j)}(t)] &\leq S_i(0)\beta \sum_{(u,i) \in E, u \neq j} E[I_{(u,i)}(t)] - \delta_i E[I_{(i,j)}(t)] \\ &= S_i(0)\beta \sum_{(u,v) \in E} A_{(i,j),(u,v)} E[I_{(u,v)}(t)] - \delta_i E[I_{(i,j)}(t)] \end{aligned}$$

Equations (4) and (6) end up looking similar in the directed edge case:

$$E[I(t)] \leq \exp((\beta A - D)t)I(0) \quad (9)$$

$$\lim_{t \rightarrow \infty} E[R(t)] = D \int_0^\infty E[I(t)]dt \quad (10)$$

### Convex Program for Directed Edges (inspired by [1])

Unlike (4) our goal isn't to directly minimize the RHS because we need the number of infected nodes not the infected edges. Equation (10) can be converted accordingly:

$$\lim_{t \rightarrow \infty} x^T E[R(t)] = x^T \int_0^\infty E[I(t)]dt$$

from which, we get the **objective**:

$$\min_D x^T D \int_0^\infty \exp((JBA - D)t)I(0)dt \quad (11)$$

given the **constraint**:

$$\sum_{i=1}^n \delta_i \leq B \quad (12)$$

## How did they perform empirically?

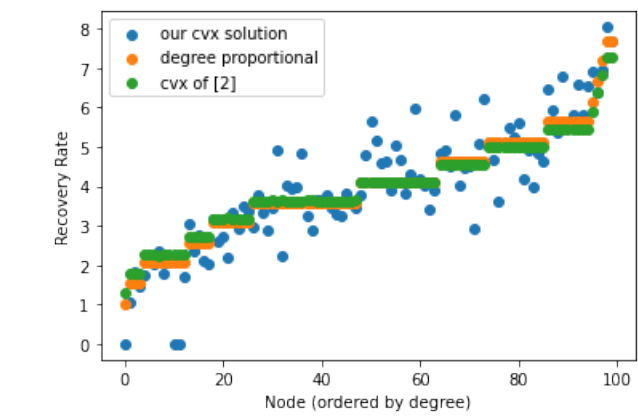


Figure 3: The different solutions for allocating antidote to ER-graph.

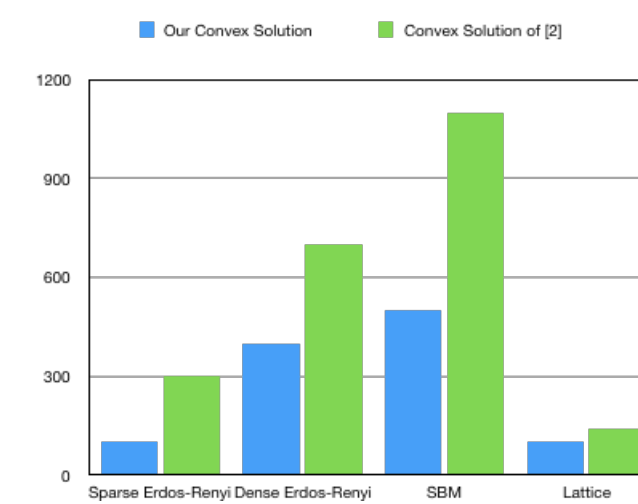
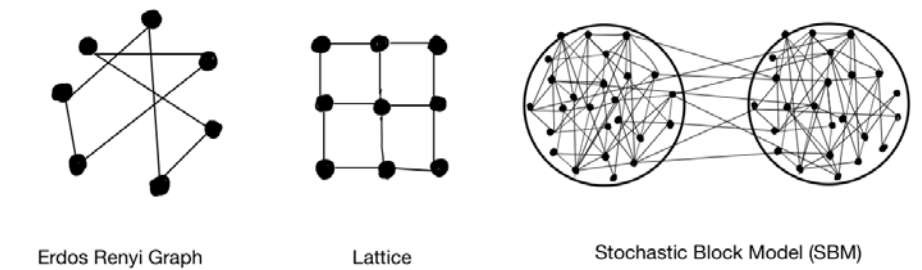


Figure 4: The budgets below which the convex programs cannot find solution for each family of graphs.



- Conjecture:** convex directed-edge always finds a solution before convex  $\leftrightarrow$  outbreak is stopped at a lower budget.
- Reasoning:** If top eigenvalue of matrix  $\beta A - D < 1$ , then the top eigenvalue of directed-edge matrix  $\beta A - D < 1$

## Next Steps

While we have found a strategy that finds a solution at lower budget than the original convex programming strategy, we still need to continue exploring strategies that can find allocations for low budgets. To do this, we plan on making the problem simpler: given a graph, which edges can we remove to stop the outbreak?

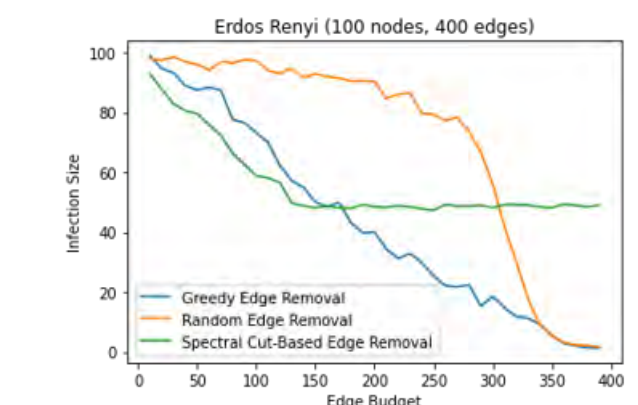


Figure 5: Different edge removal strategies.

## References

- [1] Brian Karrer, M.E.J. Newman, and Lenka Zdeborová. Percolation on sparse networks. *Physical Review Letters*, 113(20), Nov 2014.
- [2] Masaki Ogura and Victor M. Preciado. Efficient containment of exact sir markovian processes on networks. In *2016 IEEE 55th Conference on Decision and Control (CDC)*, pages 967--972, 2016.